

# LOW ACRYLONITRILE NBR PATENT LANDSCAPE REPORT

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## 1. ABSTRACT

This report evaluates the single patent in the provided manifest for relevance to low acrylonitrile nitrile butadiene rubber (NBR). The extracted evidence does not disclose an NBR synthesis patent; instead, US9267019B2 is directed to functionalized silica, wet polymer silica masterbatches, and rubber composites that may include styrene-butadiene rubber latex and, in broader formulation language, NBR as one possible rubber component. The patent's technical contribution is centered on preparing silica treated with multiple silane coupling agents, then incorporating that filler into latex-based rubber slurries and coagulating the blend to form crumb rubber composites. The disclosed process is therefore formulation- and compounding-oriented rather than a primary polymerization route for NBR.

Within the extracted evidence, the patent mentions NBR only as a usable synthetic rubber copolymer of acrylonitrile (ACN) and butadiene, and notes that nitrile content affects oil resistance and flexibility. However, no explicit acrylonitrile content, monomer feed ratio, or NBR polymerization conditions are disclosed for the target material. The examples instead detail silica functionalization with silanes, preparation of aqueous silica slurries, blending with polymer rubber latex at 70 degrees Celsius, and coagulation with coagulant. Water is clearly present in multiple downstream steps, but as a slurry, latex, and coagulation medium rather than as the core NBR polymerization medium.

Because the manifest contains only one patent and it is not centered on NBR synthesis, the report's primary finding is that no direct low-ACN NBR preparation method is captured in the extracted evidence. The patent is still relevant as a downstream compounding reference for rubber systems that may include NBR, but it does not provide a disclosed low-acrylonitrile NBR composition or synthesis route.

## 2. METHODOLOGY

### PRIMARY PATENT EVIDENCE

#### Patent 1

- Patent Number: US9267019B2
- Patent Title: Functionalized silica for silica wet masterbatches and styrene butadiene rubber ...
- Assignee: Lion Copolymer Holdings, LLC

- Jurisdiction: US
- Publication Year: 2016
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

**Polymerization / Synthesis Method** - Synthesis method: The patent discloses preparation of functionalized silica by spraying blended silanes onto dry silica in a ribbon blender, followed by heating and cooling. The treated silica is then combined with aqueous rubber latex/slurry components and coagulated to form a rubber composite; the evidence is formulation and masterbatch processing rather than NBR polymerization. - S169 silane: 0.35 pounds — Example 1 - Octyl triethoxysilane OTES: 0.35 pounds — Example 1 - Silica: 5 pounds — Example 1 - Blended silanes: 0.7 pounds — Example 1 - Acetic acid: 0.1 pounds — Example 1 - Ambient temperature: about 10 minutes — Example 1 (stirring silanes) - Ambient temperature: about 1 hour — Example 1 (tumbling in ribbon blender) - Temperature: 120 degrees Celsius — Example 1 - Heating time: 2 hours — Example 1 - Treated silica slurry: 17 weight percent treated silica in water — Example 1 - High shear blending time: about an hour — Example 1 - Treated silica slurry: 17 weight percent treated silica in water — Example 2 - Carbon black: 0.2 pounds — Example 2 - Carbon black slurry: 5.5 pounds — Example 2 - Heating temperature: 70 degrees Celsius — Example 2 - Preheated polymer rubber latex: about 17 pounds — Example 2 - Polymer rubber latex solids: 21 weight percent solids — Example 2 - Naugard™ RM51 emulsion: 0.1 pounds — Example 2 - Ergon B0300 oil emulsion: 1.3 pounds — Example 2 - Mixing time: 5 minutes — Example 2 - Mixing time: about 2 minutes — Example 2 - Coagulant addition time: about 30 seconds to about 10 minutes — Synthesis passage / Example 2 - Ribbon blender mixing rate: 5 revolutions per minute to 20 revolutions per minute — Synthesis passage - Medium & Water Role: Water is used in downstream slurry, latex, and coagulation steps; the core disclosed process is not NBR polymerization but aqueous compounding/formulation of treated silica with rubber latex. The evidence also includes a 17 wt% treated silica in water slurry and coagulation of the latex blend. - acrylonitrile content: Not disclosed in extracted evidence

**Medium & Water Role** - Water is used in downstream slurry, latex, and coagulation steps; the core disclosed process is not NBR polymerization but aqueous compounding/formulation of treated silica with rubber latex. The evidence also includes a 17 wt% treated silica in water slurry and coagulation of the latex blend. - Core reaction medium: Aqueous slurry/latex-based compounding medium; silica is first treated dry, then dispersed in water and blended with polymer rubber latex.

**acrylonitrile content** - Not disclosed in extracted evidence

**Relevant Experimental Evidence** - Example 1: Demonstrated silane blending, spraying onto silica, heating to 120 degrees Celsius, and preparing a 17 wt% silica-in-water slurry. - Example 2: Demonstrated combining treated silica slurry and carbon black slurry with preheated polymer rubber latex at 70 degrees Celsius, then coagulating the blend. - Example 2: Showed use of 21 wt% solids latex and additive emulsions (antioxidant and oil) in a latex slurry. - Example 2:

Demonstrated coagulation by slow coagulant addition to obtain coagulated crumb rubber properties.

**Technical Relevance** This patent is only indirectly relevant to low acrylonitrile NBR because it mentions NBR as one possible rubber component in broader composite formulations. It does not disclose NBR polymerization or acrylonitrile content, but it does show latex-based compounding and coagulation steps that could be used downstream of NBR production.

**PATENT COMPARISON TABLE**

| Patent | Applicant | Priority | Medium & Water Role | acrylonitrile content | Key Findings |
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| US9267019B2 | Lion Copolymer Holdings, Llc | 2016 | Water is used in downstream slurry, latex, and coagulation steps; the core disclosed process is not NBR polymerization but aqueous compounding of treated silica with rubber latex. The evidence also includes a 17 wt% treated silica in water slurry and coagulation of the latex blend. | Not disclosed in extracted evidence | Example 1: Demonstrated silane blending, spraying onto silica, heating to 120 degrees Celsius, and preparing a 17 wt% si |
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### 3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- Fewer than two selected primary patents were available, so cross-patent quantitative trends were not generated.

### 4. CONCLUSION

The single manifest patent, US9267019B2, is a rubber compounding and silica-functionalization patent rather than a direct low-acrylonitrile NBR synthesis disclosure. It mentions NBR generically but does not disclose acrylonitrile content or an NBR polymerization process, so no target-specific low-ACN NBR value can be extracted from the evidence.

## 5. REFERENCES

- US9267019B2 | Functionalized silica for silica wet masterbatches and styrene butadiene rubber ... | Lion Copolymer Holdings, Llc | US | 2016 | <https://patents.google.com/patent/US9267019B2/en>